

Dynamic and Memory-Efficient Shape-Based Methodologies for User Type Identification in Smart Grid Applications

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Abstract—Behind-the-meter (BTM) equipment identification and monitoring their availability in real time in smart grid applications require computationally efficient methodologies suitable for edge deployment. Existing shape-based approaches, whilst maintaining interpretability advantages over structure-based methods, suffer from computational bottlenecks and memory constraints when processing streaming data. This article develops three dynamic and memory-efficient updating strategies for similarity profile computation: an additive method for lossless updates, a fixed-memory approach with configurable inertia strategies, and a codebook-based technique employing dictionary learning for compressed data representation. The proposed methodologies eliminate the requirement for complete historical data reprocessing during each update, addressing critical limitations in edge computing environments. Comprehensive simulation studies using real-world photovoltaic user data demonstrate that the codebook-based approach achieves over 30% memory reduction while maintaining classification accuracy. The fixed-memory technique exhibits superior performance for applications requiring rapid change detection, with different inertia strategies providing varying sensitivity levels for diverse smart grid applications. These dynamic methodologies enable the practical deployment of interpretable BTM identification systems on resource-constrained edge devices whilst preserving the pattern recognition advantages of shape-based approaches.

Index Terms—Binary classification, data compression, data mining, dynamic updating, pattern recognition, renewable energy, time-series mining.

NOMENCLATURE

$\hat{c}_i$ Number of similar days to day i .
 $\hat{ch}_i$ Vector representing distance changes due to the updating process.

$\hat{d}_i$ Average dissimilarity of day i .
CR Vector of the compressed representation of the whole time-series data with a sequence of codewords in the codebook.
CW Vector of codewords in the codebook, indexed by k .
SP' Changes in the vector of Similarity Profile during updating, indexed by i .
SP Vector of a similarity profile.
TS Vector of metered electricity time-series data by days, indexed by day i .
 $\tilde{SP}$ Vector of a similarity profile of the old time-series data before updating.
 i Index for daily data.
 k Index for codewords.
TH Similarity threshold, i.e., day i and j are similar if $d_{i,j} \leq TH$.
 $c_{i,j}$ $c_{i,j} = 1$ if day i and j are similar. $c_{i,j} = 0$ otherwise.
 $d_{i,j}$ Dissimilarity or distance between the time series of day i and day j .
 $d_{\max}$ Maximum dissimilarity or distance between all pairs of the energy time series.
 d_{rep} Dissimilarity or distance threshold to determine if two subpatterns are interchangeable.
 l Total length of the energy time-series data.
 M Number of days of data stored in the fixed memory method.
 m Number of time intervals for one day's worth of data.
 N Number of days in the whole energy time-series data.
 W Number of the codewords in the Codebook-based method.

I. INTRODUCTION

IN MODERN power systems, residential consumers equipped with rooftop photovoltaic (PV) systems, electric vehicles, and home batteries (known as prosumers) have changed the requirements for planning and operation. Traditional load patterns and volumes have changed drastically with the integration of behind-the-meter (BTM) generation and storage systems. In addition, prosumers who participate in demand response (DR) programs (such as virtual power plants) are dynamically contributing to local or wholesale markets and actively participating

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in the management of distribution grids. Consequently, it is imperative for various smart grid stakeholders, such as aggregators, retailers, distribution network operators, and market operators, among others, to have near real-time insights into the availability of these BTM resources.

For example, knowing how many prosumers under the jurisdiction of a retailer company currently have operational rooftop solar systems would improve aggregate demand estimation and allow for more accurate forecasts for grid operations. Similarly, distribution network operators require real-time insights into BTM resource availability to predict reverse power flows and implement appropriate voltage regulation strategies, and recently estimate dynamic import and export limits. Virtual power plant aggregators depend on accurate prosumer identification for optimal dispatch decisions and DR program management. Although some countries maintain records of rooftop solar systems, documentation of other major BTM resources, such as electric vehicles, battery storage systems, and heat pumps is lacking [1]. Furthermore, the operational status of these resources could change due to equipment failures, maintenance, or behavioral modifications that require frequent updates. This underscores the necessity for developing BTM detection algorithms capable of operating effectively on low-resolution data using edge computing resources, for example existing smart meters, while preserving user privacy and minimizing communication overhead.

Current studies on BTM equipment identification methods can be categorized into two main aspects: their algorithmic approach and processing architecture [2], [3], [4]. From an algorithmic perspective, existing solutions fall further into structure-based and shape-based categories. Structure-based methods, including symbolic aggregation (SAX) and deep neural networks (DNNs), employ statistical transformations, or nontransparent modeling that sacrifice interpretability for computational scalability. In contrast, shape-based approaches, such as motif discovery analyze pattern similarity, preserving interpretability, which is a crucial feature for energy retailers and consumers, at the expense of higher computational complexity, particularly for extended time-series data. Recent advances in pattern similarity computation have made shape-based methods increasingly viable for practical deployment.

Architecturally, these methods can be implemented using centralized or distributed processing paradigms [5]. Centralized approaches require full data aggregation at a central server, creating privacy concerns, and storage burdens. Distributed implementations, such as federated learning schemes, address these issues by performing computations at edge nodes, but must remain structure-based and, thus, inherit their interpretability limitations. Furthermore, while distributed methods reduce the transmission of raw data, they still require substantial local computer power and communication bandwidth to exchange model parameters. The interpretable refined motifs in binary classification (IRMAC) framework introduced by Yuan et al. [6] is a significant progress in distributed, shape-based solutions by extracting local pattern features and ensuring interpretability with a transparent classification method. However, all existing methods, including IRMAC require complete reprocessing of historical data during each

update, which is a computationally prohibitive requirement for resource, constrained edge devices handling continuous data streams. This lack of incremental adaptability remains a critical bottleneck. This fundamental limitation motivates this article.

In this article, we present a novel framework, shown in Fig. 1, which combines the interpretability advantages of shape-based methods with a suitable dynamic computation for edge deployment. Within the proposed framework, three novel dynamic updating techniques are developed by the authors that operate on streaming data without full historical reprocessing: an additive method for lossless updates, a fixed-memory method with configurable inertia strategies, and a codebook-based technique with dynamic distributed data storage. These solutions are fast, scalable, interpretable, and flexible. To validate the effectiveness of the proposed methods, we applied them to dynamically identify rooftop PV users and assessed their reaction to behavioral changes. Extensive simulation results across diverse prosumer scenarios demonstrate that these methods reduce memory and computation requirements by 30% to 50% compared to distributed computing solutions while maintaining equivalent accuracy. Thus, this article establishes a new state-of-the-art for a practical, interpretable solution for BTM identification in edge computing environments. The novelty of this work includes the following.

- 1) Development of three dynamic updating methods (additive, fixed-memory, codebook-based) to reduce memory and computation requirements.
- 2) Comparison of different dropping strategies (low/medium/high-inertia) for the fixed-memory method, and compressing strategies for the codebook method, enabling tradeoffs between short-term awareness and long-term stability, and privacy preservation for users altering their BTM applications.
- 3) Using real-world PV data and synthetic benchmarks, demonstrating superior performance under dynamic prosumer behavior (see Section V).
- 4) Thorough performance analysis on identification accuracy, memory savings, and sensitivity on users' behavior changes for the proposed updating methods (see Section IV).

The rest of this article is organized as follows. Section II briefly explains the definitions of time series similarity measures, previous work on shape-based patterns mining, and the problem we are trying to solve in this article. Section III details the three methods proposed in this article for dynamic updating. The simulation studies are reported in Section IV, and the results are analyzed in detail. Finally, Section V concludes this article with a brief outline of prospective research directions.

II. RELATED WORK

Identification of BTM equipment has become a critical focus in the smart grid era, where related research primarily addresses two challenges: distinguishing between users equipped with different BTM devices and tracking changes in device status or usage patterns over time. Shape-based solutions combine

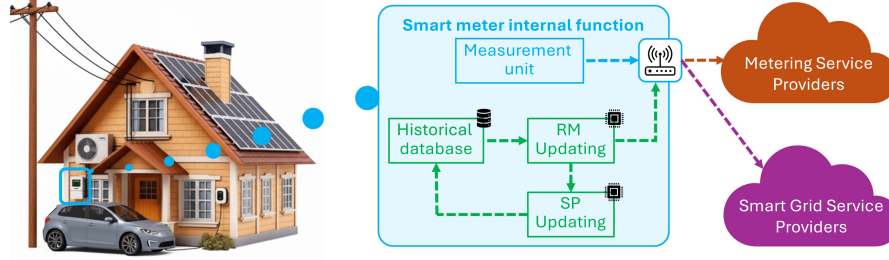


Fig. 1. High-level block diagram of the proposed methods showing a residential microgrid with smart meter and the existing and new functionalities.

	Day 1	Day 2	...	Day N
Day 1		$\langle c_{12}, d_{12} \rangle$	...	$\langle c_{1N}, d_{1N} \rangle$
Day 2	$\langle c_{21}, d_{21} \rangle$		...	$\langle c_{2N}, d_{2N} \rangle$
...	...	...	...	...
Day N	$\langle c_{N1}, d_{N1} \rangle$	$\langle c_{N2}, d_{N2} \rangle$	...	

⇓

	Day 1	Day 2	...	Day N
SP_i	SP_1	SP_2	...	SP_N

Fig. 2. Proposed SP in [6].

power system domain knowledge with data mining techniques, particularly pattern recognition and feature extraction, to uncover BTM usage features from long-term metered data [7], [8]. The following sections summarize the key concepts and methodologies that underpin this work.

A. Time Series Similarity Measures

Given an energy time series $E \in \mathbb{R}^l$, with length l , i.e., $e_l \in \mathbb{R} : E = (e_1, e_2, \dots, e_l)$, the time-series partition $E_n^m = (e_{n1}, e_{n2}, \dots, e_{nm})$ is a subpattern of E with a length of m . Given the context of residential energy usage, the length m in this article refers to the length of a day, and n is the index of n th day.

When searching for subpatterns, we need *distance metric* to distinguish similar patterns from dissimilar ones. The dissimilarity between a pair of subpatterns E_i^m and E_j^m is presented by $d_{i,j}$, which can be calculated by the $L1$ norm, Euclidean distance (ED) or dynamic time warping (DTW), among other distance metrics [7], [8]. Considering the characteristics of residential electricity consumption, time shifting, and pattern distortion due to irregular consumer behavior, and to emphasize particular temporal patterns, we use annotated DTW as our distance metric [6]. This process of retrieving subpatterns from long time series with *distance metric* is termed *similarity joins* [5], with applications in motif discovery.

B. Motif Discovery and Matrix Profile

Motif is defined as the closest subpattern pair of the time series, minimizing the *distance* [9]. The process of *motif discovery* applies *nearest-neighbor queries* to retrieve the closest neighbors

of a given subpattern E_i^m , from the long energy time series E , i.e., $R_{1NN_q} = \min(d(E_i^m, E_j^m))$ for $\{E_i^m, E_j^m\} \in E$.

In 2016, a *nearest-neighbor query*-based technique in [10], known as *MP*, was proposed that included the calculation of the z -normalized ED with the fast Fourier transform to significantly decrease the spatial and temporal complexity of the motif discovery problem. *MP* retrieves the closest neighbors for each pair of subpatterns with nearest-neighbor queries, and record the distance, i.e., $MP[i] = R_{1NN_i}$, for $i \in [1, 2, 3, \dots, N]$.

C. SPs and Refined Motifs

In smart grid applications, it is commonly required to identify a single subpattern that is repeated the most frequently and exhibits the minimum distance from other subpatterns, rather than simply identifying the nearest pair of subpatterns. Therefore, the authors proposed a new definition of motif, called *refined motif (RM)*, in [6]. Compared to extracting the closest subpatterns with *nearest-neighbor queries*, RM uses *range queries* to calculate how many subpatterns are similar. Given a subpattern $E_n^m = (e_{n1}, e_{n2}, \dots, e_{nm})$, a full TS E , a distance measure d , and a threshold ε , we can find the set of subpatterns R in E that are within the distance ε from q , i.e., $R_{RQ_{E_n^m}} = \{E_i \in E | d(E_n^m, E_i) \leq \varepsilon\}$. This process is also called *similarity self-join*. The RM can be subsequently determined using the *SP*. This involves retrieving range queries for each pair of subpatterns and noting the number of matching patterns c_i for each query subpattern. In addition, it requires recording the maximum global distance $d_{\max}$ and the average distance for each range query $\hat{d}_i$. Consequently, SP forms a vector that represents the incidences of each subpattern throughout the entire series. It can be presented as in Fig. 2 and (1), where $d_{i,j}$ is the annotated DTW distance between day i and day j , $c_{i,j}$ is a binary variable representing whether the two days are similar, the total similarity indices $\hat{c}_i$, which are integers representing the number of similar patterns with the current day i , and the average annotated DTW distance $\hat{d}_i$ normalized by $\max(d_{i,j})$ as the secondary impact factor for SP. It will differentiate the days with the same similarity indices

$$SP_i = \hat{c}_i - \frac{\hat{d}_i}{\max(d_{i,j})}. \quad (1)$$

Unlike MP assuming no domain knowledge of the data, SP takes into account domain knowledge to provide an interpretable and accurate solution for the identification of BTM appliances,

as explained in [6]. As SP computation operates at the end of users, Yuan et al. [6] introduced a distributed methodology named IRMAC to identify users with rooftop PV systems and those using electric heating systems. Simulation studies showed that IRMAC outperforms many intuitive and complex classification methods. For additional information, readers may consult [6].

D. Limitations in the Existing Approaches

Although IRMAC and other distributed approaches, such as MP and federated learning, offer clear benefits, they encounter a significant drawback: every update necessitates having access to the entire historical dataset and requires reprocessing. This presents unresolved issues for the deployment of these methods at the edge, such as computational bottlenecks, memory overflows, and data privacy concerns.

In order to tackle these challenges, we suggest a range of solutions for streaming SP updates that remove the requirement for complete historical data and reprocessing, while integrating different levels of inertia-aware edge computing along with advantages in privacy-preserving compression. These methods are optimized to perform effectively under constrained computational power, storage, and communication bandwidth, and are validated using real-world smart meter data from rooftop PV households.

III. PROPOSED DYNAMIC UPDATE METHODS

The block diagram in Fig. 1 presents the high-level architecture for the proposed dynamic update framework within typical Australian smart meter configurations. In these systems, half-hour consumption data is collected and transmitted daily to metering service providers, with retailers accessing the data for billing purposes.

The proposed framework could operate within existing smart meters or dedicated edge devices, using the availability of the measurement unit's daily data to trigger SP updates. Rather than reprocessing complete historical datasets, as required by existing methods, our approach updates the end-user's SP through single-iteration calculations using new daily data. This significantly reduces memory and computational demands, decreases the risk of privacy breaching of full historical data, while enabling real-time RM extraction and communication to smart grid service providers, as shown in Fig. 1.

This article develops three novel dynamic updating methodologies within the proposed framework specifically designed for edge-based SP computation: the additive method provides lossless updates with increasing memory requirements; the fixed-memory method maintains constant memory allocation through configurable data retention strategies; and the codebook-based method employs dictionary learning for compressed data representation with dynamic memory allocation. These methods address different operational requirements and resource constraints, making them suitable for various smart grid applications.

The additive method ensures complete accuracy by preserving all historical information, although memory consumption

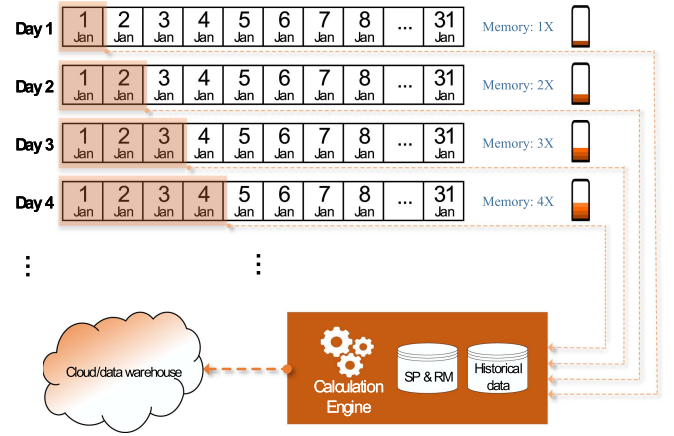


Fig. 3. Block diagram showing the operation of the additive method and its memory requirements.

increases linearly with time. The fixed-memory method offers three inertia strategies—low, medium, and high—that provide varying levels of sensitivity to detect behavioral changes while maintaining fixed computational complexity and memory requirement. The codebook-based method achieves optimal long-term performance by balancing accuracy with memory efficiency through adaptive compression techniques, as demonstrated in simulation studies.

The detailed implementation, performance characteristics, and application scenarios of each method are presented in the following sections.

A. RM Update With Additive Method

Using January as an example, the additive method provides a baseline approach for dynamic SP, represented schematically in Fig. 3. The daily profile for each day requires a fixed memory allocation of $1X$ bytes, where X is the memory requirement for the demand profile for one day. The required memory grows daily.

The incoming data is compared with all existing historical patterns to update $\vec{SP}$ with changes $\vec{SP}'$. Therefore, the updated time series contains all the daily data from day 1 to the day before. After updating, the new time series and SP will be stored at the user's end, where the RM can be updated from the new SP, as shown in Fig. 2. It can be seen in the figure that the memory requirement will grow constantly over time. Equations (2)–(5) present the additive updating process from day N to day $N + 1$ in three steps. In the first step, the global maximum dissimilarity $d_{\max}$ is calculated with the distance between day $N + 1$ and days 1 to N . Second, changes in SP, that is, SP'_i , are calculated based on the global maximum dissimilarity, and the counters are updated by the dissimilarity. Third, SP is calculated by updating the old SP value, $\tilde{SP}_i$, with the new incoming value, SP_{N+1} as in (4) and (5). The procedure takes one iteration to update the RM with linear complexity

$$\max(d_{i,j}) = \max(\tilde{d}_{\max}, d_{i,N+1}), \quad 1 \leq i \leq N \quad (2)$$

Algorithm 1: Additive Method for Updating SP.

Require: TS contains N days data $TS_{[1:N]}$ with a new incoming data TS_{N+1} to be updated, SP with N days' record $SP_{[1:N]}$, a maximum dis-similarity $d_{\max}$, and a threshold TH

$c_{[1:N]} \leftarrow [0, 0, \dots, 0]$ $\triangleright$ initialise counters
 $ch_{[1:N]} \leftarrow [0, 0, \dots, 0]$ $\triangleright$ initialise changes of distances
 $\hat{d}_{N+1} \leftarrow 0$ $\triangleright$ initialise dissimilarity for new data

for $i = 1, 2 \dots N$ **do**
 $d_{i,N+1} \leftarrow DTW(TS_{N+1}, TS_i)$
 $\hat{d}_{N+1} \leftarrow \hat{d}_{N+1} + d_{i,N+1}$
 $d'_{\max} \leftarrow \max(d_{\max}, d_{i,N+1})$ $\triangleright$ update max distance
 $ch_i \leftarrow d_{i,N+1}$
if $d_{i,N+1} \leq TH$ **then** $\triangleright$ check threshold
 $c_i \leftarrow c_i + 1$
 $c_{N+1} \leftarrow c_{N+1} + 1$
end if
end for

$avg_{scaled} \leftarrow \lceil SP_{[1:N]} \rceil - SP_{[1:N]}$
 $avg' \leftarrow \frac{avg_{scaled} \cdot (N-1) \cdot d_{\max} + ch_{[1:N]}}{d'_{\max} \cdot N} - avg_{scaled}$
 $SP_{[1:N]} \leftarrow SP_{[1:N]} + c_{[1:N]} - avg'$ $\triangleright$ Equation (3)–(4)
 $SP_{N+1} \leftarrow c_{N+1} - \frac{d_{new} \cdot (N+1)}{d'_{\max} \cdot N}$ $\triangleright$ Equation (5)
return SP, TS, $d'_{\max}$ $\triangleright$ RM is TS_i where $SP_i = \max(SP)$

$$SP'_i = c_{i,N+1} - \frac{d_{i,N+1} \cdot \tilde{d}_{\max}}{\max(d_{i,j})} + \frac{(\lceil SP_i \rceil - \tilde{SP}_i) \cdot (N-1)}{N}$$

$$1 \leq i \leq N \quad (3)$$

$$SP_{[1:N]} = SP'_{[1:N]} + \tilde{SP}_{[1:N]} \quad (4)$$

$$SP_{N+1} = \sum_{i=1}^N c_{i,N+1} - \frac{\sum_{i=1}^N d_{i,N+1}}{\max(d_{i,j}) \cdot N} \quad (5)$$

Algorithm 1 shows a step-by-step computation process. For a time series with length l and window size m , which grows daily, recomputing the SP takes $O(l^2m)$ time complexity, as discussed in [6]. On the other hand, updating an incoming day's data at the user's end with the additive method on the same data length takes the time complexity of $O(lm)$.

The main bottleneck of the additive method is that memory and computational time requirements increase with the number of days, as shown in Fig. 3. In addition, the additive method provides a global RM because of using the entire historical data. Therefore, it can be as accurate as the method proposed in [6]. However, this method is less sensitive to identifying the user's type switching, i.e., a non-PV user switching to a PV user or vice versa. To address these issues, we propose a modified updating approach in the next section.

B. RM Update With Fixed-Memory Method

The fixed-memory method addresses the unbounded memory growth of the additive approach by maintaining a sliding window of the new incoming subpattern, as shown in Fig. 4.

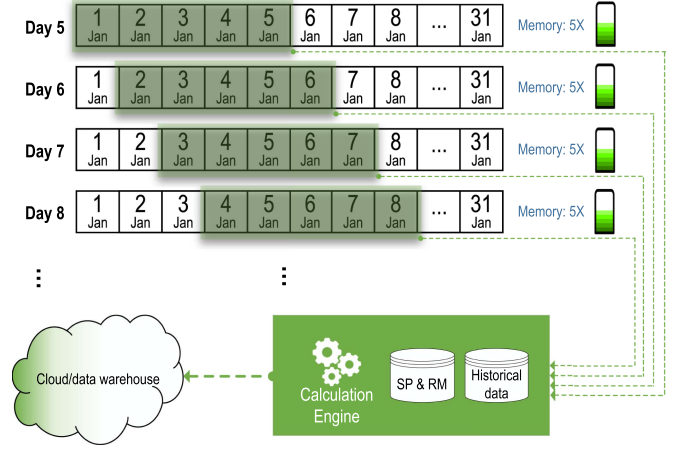


Fig. 4. Block diagram representing the operation of the fixed-memory method and its memory requirement.

New daily data replaces data from a selected past day in the memory, thereby maintaining a consistent memory allocation size, denoted M . Compared to the additive method, the time complexity of the fixed-memory method is $O(M \cdot m)$, which is l/M times the complexity of the additive method since l is normally much larger than M . Algorithm 2 illustrates the sequential procedure of the fixed-memory method.

This method is a lossy solution in terms of preserving the actual SP values due to the method tracking the maximum distance within the memory as opposed to the global maximum distance, i.e., $\max(d_{i,j})$ in (2). The extracted RM is expected to be accurate because the decimal relationship in the SP is preserved, with its decimal part scaled at the same level in (5). There are two hyperparameters in the fixed-memory method, i.e., subpattern dropping strategy and the window size that are discussed next.

1) Different Dropping Strategies: The performance of the fixed-memory method is critically dependent on the strategy to select TS_{out} . Three dropping strategies are evaluated, which could be used in different smart grid applications as follows.

- Low-inertia strategy:** In this approach, we remove the oldest day profile from memory to save the incoming daily data. In this way, we can access the latest M days of consumer demand profiles. As a result, this strategy is heavily influenced by the most recent historical data and the temporal dynamics of it will play the main role in finding RM. In other words, it offers rapid detection of behavior changes, but potentially discards valuable historical patterns.
- High-inertia strategy:** As daily time series and SP are maintained and do not have to be sequential, we suggest implementing a “high-inertia strategy” that eliminates data with minimal SP value to maintain long-term memory. This strategy will preserve similar patterns in the memory. Therefore, it preserves dominant motifs at the cost of reduced sensitivity to subtle changes.
- Medium-inertia strategy:** Taking into account the issues of the previous reduction strategies, we propose the third

Algorithm 2: Fixed-Memory Method.

Require: Variable TS contains M stored daily subpatterns $TS_{[1:M]}$ with an incoming day's data TS_{M+1} to update, variable SP with M days' similarity information $SP_{[1:M]}$, a maximum dis-similarity $d_{\max}$, and a threshold TH

$TS_{in} \leftarrow TS_{M+1}$ $\triangleright$ new incoming data
 $TS_{out} \leftarrow TS_{index}$ $\triangleright$ drop data by index
 Remove TS_{index} from TS
 $c_{[1:M]} \leftarrow [0, 0, \dots, 0]$ $\triangleright$ initialise counters
 $ch_{[1:M]} \leftarrow [0, 0, \dots, 0]$ $\triangleright$ initialise changes of distances
 $d_{new} \leftarrow 0$ $\triangleright$ initialise dissimilarity for new data
for $i = 1, 2, 3 \dots M-1$ **do**
 $d_{out} \leftarrow DTW(TS_{out}, TS_{[i]})$
 $d_{in} \leftarrow DTW(TS_{in}, TS_{[i]})$
 $d_{new} \leftarrow d_{new} + d_{in}$
 $d'_{\max} \leftarrow \max(d_{\max}, d_{in})$ $\triangleright$ update distance
 $ch_i \leftarrow d_{in} - d_{out}$
 $ch_i \leftarrow \frac{ch_i}{M}$ $\triangleright$ changes in average
 if $d_{out} \leq TH$ **then** $\triangleright$ check threshold
 $c_{[i]} \leftarrow c_{[i]} - 1$
 end if
 if $d_{in} \leq TH$ **then** $\triangleright$ check threshold
 $c_i \leftarrow c_i + 1$
 $c_M \leftarrow c_{M+1} + 1$
 end if
end for
 $avg_{scaled} \leftarrow \lceil SP_{[1:M-1]} \rceil - SP_{[1:M-1]}$
 $avg' \leftarrow \frac{avg_{scaled} \cdot d'_{\max} + ch_{[1:M-1]}}{d'_{\max}} - avg_{scaled}$
 $SP_{[1:M-1]} \leftarrow SP_{[1:M-1]} + c_{[1:M-1]} - avg'$
 $SP_M \leftarrow c_M - \frac{d_{new}}{d'_{\max}}$
return SP, TS, $d'_{\max}$ $\triangleright$ RM is TS_i where $SP_i = \max(SP)$

strategy in which we eliminate daily data with the median SP value. The idea is to keep the most similar and dissimilar patterns in the database. In this way, the medium-inertia strategy maintains diversity in the stored patterns while filtering outliers.

To better explain the three strategies, consider an example in which a consumer's PV solar system malfunctions, resulting in a temporary shift from a solar user to a nonsolar user, as shown in Fig. 5. The yellow and light-blue areas in the figure represent the periods before and after the breakdown of the solar system. It can be seen that it takes three and four updates for the low-inertia and medium-inertia strategies, respectively, to identify the breakdown (shown by the vertical line). However, the high-inertia strategy cannot detect this change even after four updates. Therefore, it can be concluded that the low-inertia strategy method is good at tracking even the smallest changes in user behavior. In contrast, the high-inertia strategy works best for users with stable behavior. A comprehensive simulation study on the capabilities of the three methods is presented in Section IV.

2) Hyperparameter Justification for Memory Size: The performance of the fixed-memory method depends critically on the parameter M , which determines how many daily subpatterns are

preserved in memory. Although larger M retains more historical patterns at increased memory and computation costs, smaller values improve efficiency but risk losing important temporal features. This tradeoff is particularly relevant for edge devices with limited computational resources and storage. To guide practitioners in selecting M , Section IV provides a systematic analysis that examines the following:

- accuracy versus memory tradeoff for different values of M ;
- identification of the knee point where accuracy gains diminish;
- practical recommendations based on application requirements.

C. RM Update With Codebook-Based Method

The codebook-based method provides an adaptive solution for dynamic RM updates by employing dictionary learning techniques to achieve memory efficiency while preserving pattern fidelity [5], [11]. This method is especially appropriate for long-duration applications where memory limitations are significant, but it is necessary to preserve full historical data in a compressed format. The method maintains two key data structures, namely codeword and representations. The codeword is a subpattern cw_i in the time series representing a set of other similar subpattern neighbors whose distance is smaller than a replaceable distance threshold d_{rep} , that is, $\{E_i \in E | d(cw_i, E_i) \leq d_{rep}\}$. Representation cr_i is the pointer that maps the compressed time series partition i to the codeword cw_s of the codebook, that is, $E_i' = D \cdot cr_i = cw_s$. We adopt the same idea for the real-time updating problem of RM discovery, as shown in the block diagram of Fig. 6. Since the dissimilarity of the daily patterns in our proposed dynamic RM is measured independently, the RM discovery process does not require the subpatterns to be in a temporal sequence. Consequently, we propose two substrategies for the codebook method, which are discussed in the following sections.

1) Codebook Method With Compressed Representation:

The first strategy is the conventional codebook method. In this method, the codebook and the historical SP are stored at the user's end. The distance between the new incoming data and the existing codewords in the codebook is computed. If the distance of the new incoming data and the most similar codeword is smaller than a predefined threshold, it will be replaced with an existing codeword. Otherwise, it will be added to the codebook as a new codeword. In general, this method will compress the end-user's data with a codebook and recover the data with compressed representation before the updating process. The pseudocode for this approach is presented in Algorithm 3.

2) Codebook Without Compressed Representation:

In contrast to the earlier codebook approach, we introduce a novel method that presumes reconstructing the original time series from codewords is not required. Instead, an alternative codebook known as the patterns dictionary (PD) is stored at the end-user's location. PD is proposed to store the codewords as keys and the occurrences as values. The end user can maintain and revise the complete SP based on codewords and occurrences, though

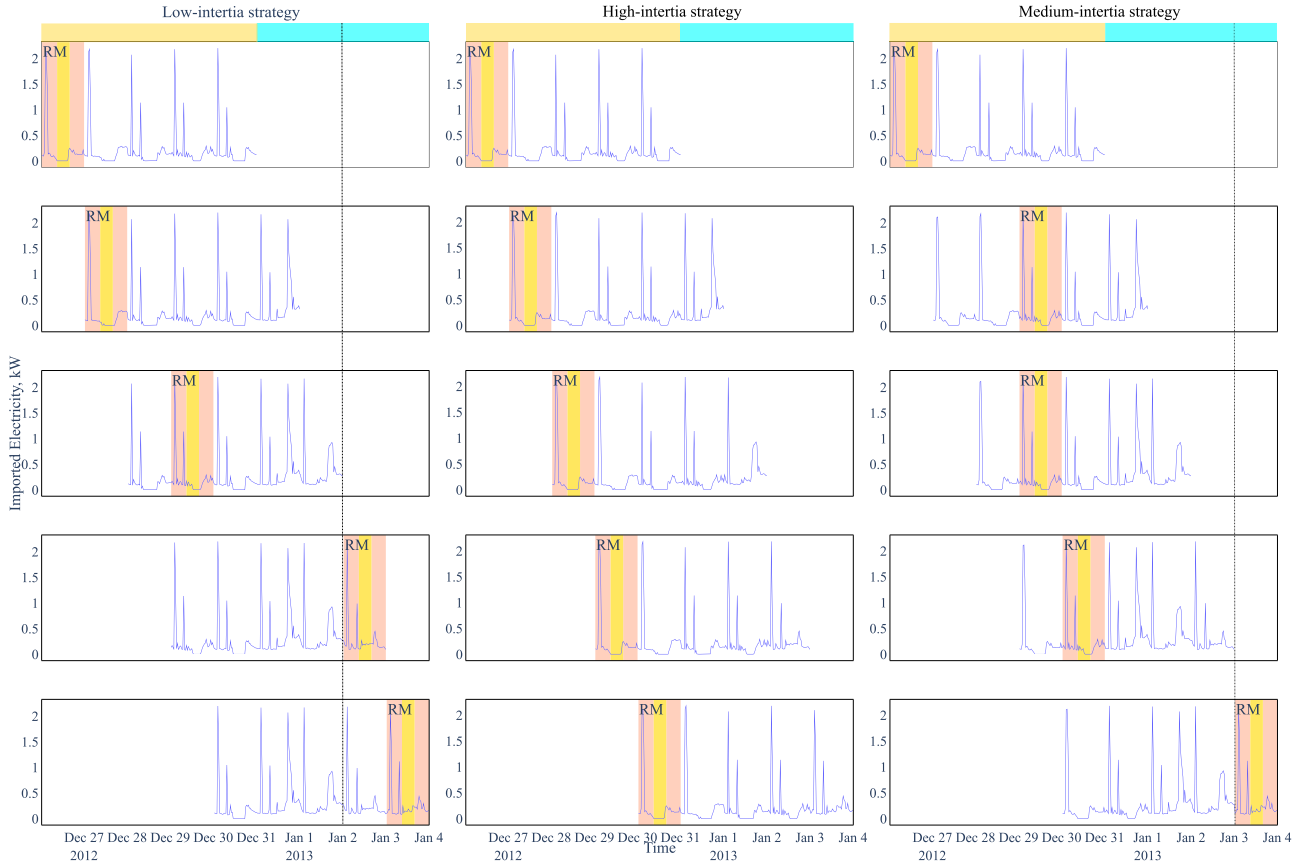


Fig. 5. Comparison of three variations of the fixed-memory method in detecting solar type switching, window size = 5.

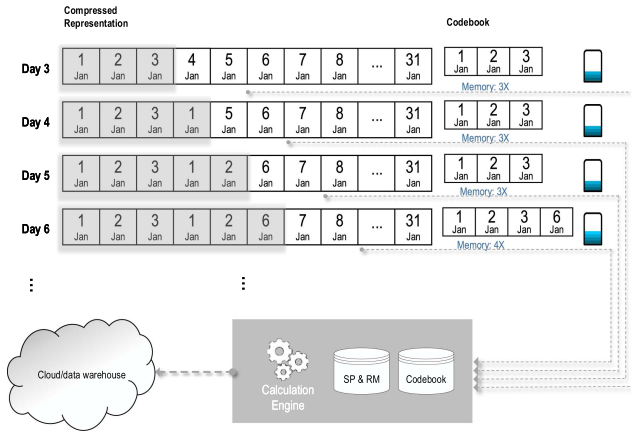


Fig. 6. Block diagram illustrating the Codebook method for dynamic RM discovery.

it lacks a temporal aspect since the SP's index values do not indicate collection times. Compared to the Codebook model with compressed representation, this strategy preserves the relationship of each subpattern but in an anonymous way because the temporal sequences are not included. More specifically, the SP's subpattern with a higher index does not necessarily refer to newer data than the subpattern with a lower index. Furthermore, it requires less memory since we do not need a compressed

representation. The implementation of this strategy is explained step by step in Algorithm 4.

IV. SIMULATION STUDIES

To assess the performance of our proposed methods, the solar home data from New South Wales, Australia, is used for the simulation studies, which contains half-hourly PV generation and demand data from 300 residential consumers, spanning three continuous years from 2010–2011 to 2012–2013 [12]. Our evaluation focuses on the following three key aspects:

- 1) PV user identification accuracy;
- 2) computational efficiency;
- 3) memory requirements.

Since both the fixed-memory and codebook-based methods have different substrategies (see Sections III-B and III-C), we report and discuss the results of the fixed-memory and codebook models individually to better show their advantages and limitations. Finally, the overall accuracy of all the proposed methods is compared in Section IV-C for different data lengths.

A. Fixed-Memory Method: Evaluation on Dropping Strategies and Memory Sizes

In order to thoroughly assess the fixed-memory approaches, a comprehensive simulation study was carried out where users transitioned between being solar and nonsolar users. We selected

Algorithm 3: Codebook With Compressed Representation.

Require: Variable CW contains W stored codewords
 CW_[1:W] with an incoming TS_{in} to update, variable SP with N days' similarity SP_[1:N], a compressed representation with N day's code number CR_[1:N+1], a maximum dis-similarity $d_{\max}$, and a distance value which is seen as replaceable d_{rep}

$c_{[1:N]} \leftarrow [0, 0, \dots, 0]$ $\triangleright$ initialise counters
 $ch_{[1:N]} \leftarrow [0, 0, \dots, 0]$ $\triangleright$ initialise changes of distances
 $d_{new} \leftarrow 0$ $\triangleright$ initialise dissimilarity for the new data
 $d_{min} \leftarrow d_{rep}$
 $TS_{[1:N]} \leftarrow CW_{CR_{[1:N]}}$ $\triangleright$ recover TS
 $CR_{N+1} \leftarrow W + 1$ $\triangleright$ initialise codebook

for $i = 1, 2, 3 \dots N$ **do**
 $d_{in} \leftarrow DTW(TS_{in}, TS_i)$
 $d_{new} \leftarrow d_{new} + d_{in}$
 $d'_{\max} \leftarrow \max(d_{\max}, d_{in})$ $\triangleright$ update d
 $ch_{[i]} \leftarrow d_{in}$
if $d_{in} \leq TH$ **then** $\triangleright$ check threshold
 $c_i \leftarrow c_i + 1$
 $c_{N+1} \leftarrow c_{N+1} + 1$
end if
if $d_{in} \leq d_{min}$ **then** $\triangleright$ check if replaceable
 $d_{min} \leftarrow d_{in}$
 $CR_{N+1} \leftarrow CR_i$
end if
end for

$avg_{scaled} \leftarrow \lceil SP_{[1:N]} \rceil - SP_{[1:N]}$
 $avg' \leftarrow \frac{avg_{scaled} \cdot (N-1) \cdot d_{\max} + ch_{[1:N]}}{d'_{\max} \cdot N} - avg_{scaled}$
 $SP_{[1:N]} \leftarrow SP_{[1:N]} + c_{[1:N]} - avg'$
 $SP_{N+1} \leftarrow c_{N+1} - \frac{d_{new}}{d'_{\max}}$
if $CB_{N+1} == W + 1$ **then**
 $CR_{W+1} \leftarrow TS_{in}$
end if
return SP, CW, $d'_{\max}$, CR $\triangleright$ RM is CW_{CR_i} where
 $SP_i == \max(SP)$

100 random users with rooftop solar systems from the dataset, assuming their systems would fail on day 10, making them non-solar users afterward. Also, another dataset with 100 nonsolar users was created, where they acquired solar systems on day 10, resulting in their conversion to solar users for the remaining duration of the study. The frequency of daily RM updates was tracked to observe the performance of different methods to detect user transitions. Figs. 7 and 8 illustrate the results for solar users becoming nonsolar users and vice versa.

The simulations reveal distinct performance characteristics across the three inertia strategies in tracking user-type transitions. The low-inertia strategy shows remarkable responsiveness, achieving 100% detection of transitions between solar and nonsolar states. Conversely, the high-inertia strategy effectively retains patterns but poorly detects changes, missing 36% of nonsolar-to-solar and 57% of solar-to-nonsolar

Algorithm 4: Codebook Method Without Compressed Representation.

Require: A dictionary with W stored daily patterns as keys and occurrences as values, i.e.,
 $PD = \{CR_1 : n_1, \dots, CR_W : n_W\}$, an incoming TS_{in} to update, variable SP with N days SP_[1:N], a maximum dis-similarity $d_{\max}$, and a value of distance which can be seen as predefined d_{rep}

$c_{[1:N+1]} \leftarrow [0, 0, \dots, 0]$ $\triangleright$ initialise counters
 $ch_{[1:N]} \leftarrow [0, 0, \dots, 0]$ $\triangleright$ initialise changes of distances
 $d_{new} \leftarrow 0$ $\triangleright$ initialise dissimilarity for new data
 $d_{min} \leftarrow d_{rep}$
 $p \leftarrow W + 1$ $\triangleright$ initialise the position for new data in PD
 $cr \leftarrow 1$ $\triangleright$ track the position in SP

for $i = 1, 2, 3 \dots W$ **do**
 $d_{in} \leftarrow DTW(TS_{in}, CR_i)$
 $n_i \leftarrow PD_{CR_i}$ $\triangleright$ occurrence of the data
 $d_{new} \leftarrow d_{new} + d_{in} \cdot n_i$
 $d'_{\max} \leftarrow \max(d_{\max}, d_{in})$ $\triangleright$ update d
 $ch_{[cr:cr+n_i]} \leftarrow d_{in}$
if $d_{in} \leq TH$ **then** $\triangleright$ check threshold
 $c_{[cr:cr+n_i]} \leftarrow c_{[cr:cr+n_i]} + 1$
 $c_{N+1} \leftarrow c_{N+1} + 1$
end if
if $d_{in} \leq d_{min}$ **then** $\triangleright$ check if replaceable
 $d_{min} \leftarrow d_{in}$
 $p \leftarrow i$
end if
 $cr \leftarrow cr + n_i$ $\triangleright$ update the position

end for

$avg_{scaled} \leftarrow \lceil SP_{[1:N]} \rceil - SP_{[1:N]}$
 $avg' \leftarrow \frac{avg_{scaled} \cdot (N-1) \cdot d_{\max} + ch_{[1:N]}}{d'_{\max} \cdot N} - avg_{scaled}$
 $SP_{[1:N]} \leftarrow SP_{[1:N]} + c_{[1:N]} - avg'$
 $SP_{in} \leftarrow c_{[N+1]} - \frac{d_{new}}{d'_{\max}}$ $\triangleright$ new data's SP

if $p \leq W + 1$ **then**
 $n_p \leftarrow n_p + 1$
 $cr \leftarrow n_1 + n_2 + \dots + n_p$ $\triangleright$ find the position to insert new SP
 $SP_{[cr+1:N+1]} \leftarrow SP_{[cr:N]}$ $\triangleright$ shift one place for new data
 $SP_{[cr]} \leftarrow SP_{in}$ $\triangleright$ insert new SP

else
 $CR_{W+1} \leftarrow TS_{in}$
 $n_{W+1} \leftarrow 1$
 $SP_{N+1} \leftarrow SP_{in}$

end if

return SP, PD = {CR₁ : $n_1, \dots$ }, $d'_{\max}$ $\triangleright$ RM is CR_i where
 $SP_{[n_1+\dots+n_i]} == \max(SP)$

transitions. The medium-inertia strategy strikes a balance, detecting over 50% of transitions within 30 days while maintaining reasonable long-term stability. The differences in performance are attributed to core design variations: the sliding window method of low-inertia favours recent information, the feature-preserving strategy of high-inertia emphasizes

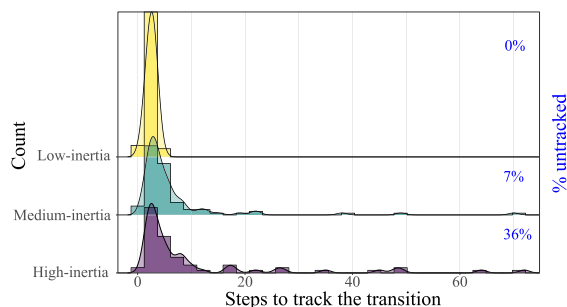


Fig. 7. Inertia study for 100 nonsolar users becoming solar users: median number of iterations to detect the change is 3, 4, and 11 for low-, medium-, and high-inertia strategies.

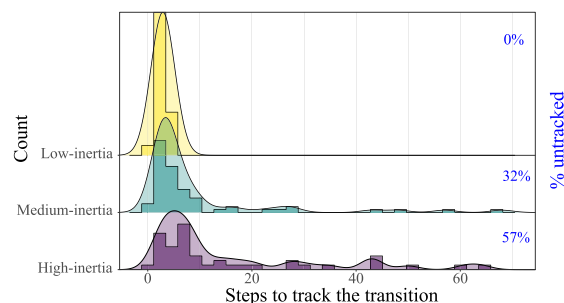


Fig. 8. Inertia study for 100 solar users becoming nonsolar users: median number of iterations to detect the change is 3, 7, and 80 for low-, medium-, and high-inertia strategies.

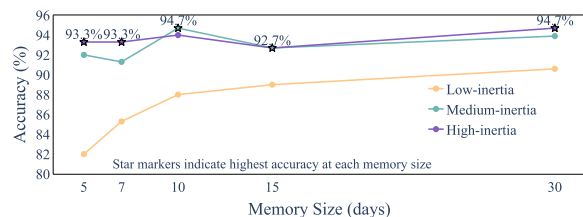


Fig. 9. Performance of the three dropping strategies.

on detecting stable patterns, and the selective retention in medium-inertia ensures the maintenance of diversity. Notably, all strategies perform better at detecting nonsolar-to-solar transitions (7%–32% unidentified) versus solar-to-nonsolar (36%–57% unidentified), as solar patterns exhibit more distinctive SP signatures that are easier to preserve in memory-limited scenarios.

We also ran another simulation study with different memory sizes to assess its impact on the performance of the fixed-memory approaches, i.e., identifying solar users from nonsolar users. The results are shown in Fig. 9. The low-inertia strategy shows strong memory dependence, with accuracy improving from 82% to 91% as memory expands from 5 to 30 days. The high-inertia strategy achieves the highest average accuracy (94%), demonstrating stable long-term feature extraction. Medium-inertia achieves peak accuracy with 10 days of memory size, validating its balanced design on tracking transitions and preserving featured patterns.

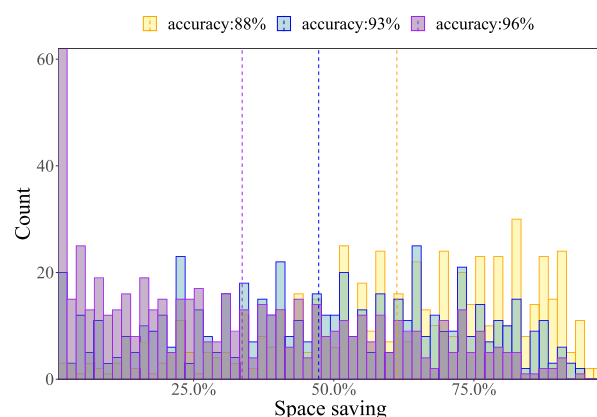


Fig. 10. Tradeoff between memory saving and accuracy in the codebook-based method for 299 solar users and 299 nonsolar users.

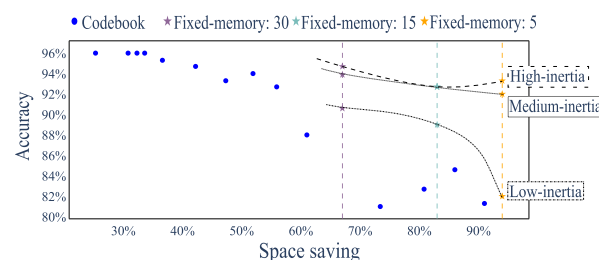


Fig. 11. Accuracy versus memory saving Pareto front for codebook and fixed-memory techniques.

TABLE I
COMPRESSION RATIO OF THE CODEBOOK METHOD

Month	Memory saving	min	max	average
1		9%	96%	30%
2		8%	98%	31%
3		6%	98%	34%

TABLE II
PERFORMANCE OF THE DIFFERENT METHODS IN THE SOLAR USER IDENTIFICATION PROBLEM

Month	Accuracy	Additive	Fixed-Memory (low)	Fixed-Memory (high)	Fixed-Memory (medium)	Codebook
1		94.0%	88.0%	92.7%	92.0%	94.0%
2		94.7%	86.0%	94.7%	96.0%	96.0%
3		96.0%	89.3%	92.7%	92.7%	96.0%

B. Codebook: Compression Rate and Accuracy

The compression performance is quantified by *memory saving*, which is the ratio between saved memory versus total memory. A sensitivity analysis is conducted for 300 consumers with 1, 2, and 3 months of data from summer. The results are summarized in Table I. It can be seen that the memory saving rate increases when more data are included. In other words, adding more data may lead to more stable codewords.

One hyperparameter in the codebook-based method is the threshold to determine if one subpattern can be replaced with the codeword. A lower threshold results in more codewords in the codebook or PD; hence, less memory saving. On the other hand, the distance between replaced subpatterns and the

TABLE III
COMPARISON OF THE FEATURES OF THE PROPOSED RM UPDATING METHOD

Method	Advantages	Disadvantages	Applications
Additive	• Lossless SP values • Simple • Preserves data	• Memory intensive • Highest time complexity	• Long-term behavioural studies with SP
Fixed-memory	• Lowest memory requirement • Fast update (low-inertia) • Feature enhancement (high-inertia) • Balanced performance (medium-inertia)	• Low accuracy • Permanent knowledge elimination (in low-inertia strategy)	• Short-term modelling • Short-term tracking of changes
Codebook-based	• Low memory requirement • Preserve the changes from historical records • Flexible • High security (no sequential information) • High compression ratio for long TS	• Cannot preserve original data • Memory intensive for short TS	• Long-term users • Sensitive data to preserve privacy

codewords will be smaller. As a result, there is a tradeoff between memory saving and accuracy. In particular, the compression performance for each user is different because of the variability in their patterns. To measure the tradeoff between accuracy and compression, we performed a simulation study in which the threshold is changed from 0.5 to 2 to compute the memory saving for the 300 users. In addition, we measured the performance of the extracted RMs by identifying solar users with a linear classifier developed in our previous work [6]. Fig. 10 illustrates the memory saving distributions for end-users with varying classification accuracy.

In general, as illustrated in Fig. 10, memory saving exhibits considerable variability between users. At 96% accuracy, 87 users barely show any memory saving. However, most user data can be compressed to a certain degree, with an average memory reduction of 30%. As accuracy declines, the distribution shifts toward greater memory savings. At 88% accuracy on the classification problem, most users achieve more than 50% memory saving.

Fig. 11 presents the Pareto front derived from additional experiments on the tradeoff between average memory savings and accuracy, using data collected over a 3-month period for 598 users. The fixed-memory method with the three elimination strategies is also included for performance comparison.

From Fig. 11, it is apparent that the codebook method without using compressed representation managed to achieve an average of 30% memory savings without compromising accuracy. At compression levels exceeding 50% memory savings, the fixed-memory method's high- and medium-inertia strategies perform substantially better. In contrast, as indicated in Table I, the memory savings in the codebook method increase with longer time series. This suggests that accuracy in the high compression range may improve with extended historical data availability.

C. Accuracy Analysis for the Proposed Updating Methods

The overall accuracy and memory size are presented in Table II, where the memory size in the fixed-memory method is 15 (i.e., $M = 15$) and the compression rate of the codebook method is 30%.

The codebook and additive methods show the overall best accuracy among the three methods, followed by the fixed-memory with high-inertia strategy. The accuracy of the fixed-memory method with a low-inertia strategy significantly varies with

different data. Based on simulation studies in previous sections, the main features of the three updating methods are summarized in Table III.

D. Summary of the Performance Analysis and Implications

The simulation results reveal different performance characteristics between methodologies. The 96% accuracy of the additive method comes at the cost of linear memory growth, making it unsuitable for long-term deployment. The fixed-memory approach shows remarkable efficiency gains, reducing computational complexity from $O(l^2 m)$ to $O(M \cdot m)$, where $M \ll l$. This represents an 85%–90% reduction in processing requirements for typical deployment scenarios.

The performance of the codebook-based method exhibits data-dependent characteristics. For users with highly repetitive patterns, compression ratios exceed 60%, while diverse consumption profiles achieve 20%–30% compression, without compromising the accuracy. This variability suggests the suitability of the method for different consumer segments, with industrial or commercial applications likely benefiting from higher compression rates due to more predictable usage patterns.

V. CONCLUSION

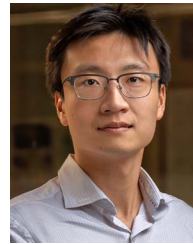
This article proposes three RM updating methods and six substrategies for smart grid applications. The performance and sensitivity of each method to its respective hyperparameters are analyzed with a set of simulation studies. In general, the additive method yields the most accurate outcomes, although with an increased demand for memory and computational resources. The fixed-memory method offers the highest memory saving and fixed temporal complexity. The three data elimination strategies are proposed that can be used in different circumstances. The low-inertia strategy is sensitive to behavioral changes and memory size; hence, it is good for tracking changes. The high-inertia strategy cannot detect sudden changes but is good at detecting similar subpatterns in daily data. The medium-inertia strategy is a balanced option in terms of accuracy, memory saving, and detecting changes. The codebook method is the most balanced solution, providing more than 30% of memory saving without sacrificing accuracy. With this method, both the codebook with compressed representation and PD provide the same RM, but the former can easily recover the whole time series while the latter is anonymous. The memory savings of the codebook method

increase with the data size, making it a more robust solution for storing long-term patterns or high-resolution data for end users. Another significant advantage of the codebook method is the privacy-preserving feature at the user's end.

The authors envision three possible directions for future work. Firstly, the memory size for the fixed-memory method and the predefined threshold for the codebook-based method were chosen based on educated guesses. These hyperparameters require systematic optimization as they are fundamentally tied to data resolution and application requirements. We plan to develop generalized selection strategies through extended sensitivity analysis across diverse datasets. Second, the extracted RM can be used in more applications, e.g., demand modeling, forecasting, and demand-side management measures. For instance, it can help detect new EV users, new air conditioners or heat pumps, or shifts between cooling and heating regimes in near real-time. It could help detect malicious behaviors, such as electricity theft. Future research could focus on broadening the application of the proposed framework to additional smart grid challenges. Third, our study is limited to the data from the utility meters that are accessible to aggregators and retailers. With the availability of high-resolution data from smart meters, combined with our memory-efficient updating strategies, such as the fixed-memory and codebook method, it becomes possible to identify a larger number of appliances, providing better insight into consumer behavior and variations without memory outage concern [13], [14].

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